

Yunshan Duan

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EMPLOYMENT

Postdoctoral Fellow, Johns Hopkins University

Aug 2025 - Present

Department of Applied Mathematics and Statistics

Advisor: Dr. Yanxun Xu

EDUCATION

University of Texas at Austin

Austin

Ph.D. in Statistics

Aug 2025

Advisor: Dr. Peter Müller

Dissertation: Bayesian nonparametric inference in bioinformatics applications

Fudan University

Shanghai

B.S. in Mathematics and Applied Mathematics

Jun 2021

RESEARCH INTERESTS

My research develops Bayesian and statistical machine learning methods that incorporate scientific structure into the analysis of complex biomedical data.

- **Methodology:** Bayesian statistics; self-supervised representation learning; Bayesian nonparametrics; Bayesian clinical trial design.
- **Applications:** health records and medication data; HIV, mental health, and cognition; single cell and spatial transcriptomics; adaptive clinical trials and surrogate endpoints.

PREPRINTS & PUBLICATIONS

- **Duan, Y., Korpela, E., & Xu, Y. (2026).** Self-Supervised Representation Learning for Longitudinal Polypharmacy Patterns. arXiv preprint arXiv:2608.06743. (*Under Review*).
- **Duan, Y., & Williamson, S. (2026).** Self-Supervised Learning with Gaussian Processes. arXiv preprint arXiv:2512.09322. (*Under Review*).
- **Duan, Y., Guo, S., Yan, H., Wang, W., Müller, P. (2025).** Spatially aligned random partition model on spatial transcriptomics data. *Bayesian Analysis*, Advance Publication 1-26, 2025.
- **Duan, Y., Guo, S., Wang, W., & Müller, P. (2025).** Immune profiling among colorectal cancer subtypes using dependent mixture models. *Journal of the American Statistical Association*, 1-23. (**Winner of 2024 ASA-SBSS student paper competition**)
- **Duan, Y., & Parast, L. (2024).** Flexible evaluation of surrogate markers with Bayesian model averaging. *Statistics in Medicine*, 43(4), 774-792.
- **Duan, Y., Yuan, S., Ji, Y., & Müller, P. (2024).** A unified decision framework for phase I dose-finding designs. *Statistics in Biosciences*, 16(1), 69-85.
- **Tec, M., Duan, Y., & Müller, P. (2023).** A comparative tutorial of Bayesian sequential design and reinforcement

learning. *The American Statistician*, 77(2), 223-233.

- **Duan, Y.**, Müller, P., & Ji, Y. (2023). Comment: Response-adaptive randomization in clinical trials: From myths to practical considerations. *Statistical Science*, 38(2), 212-215.
- Müller, P., **Duan, Y.**, & Garcia Tec, M. (2022). Simulation based sequential design. *Pharmaceutical Statistics*, 21(4), 729-739.
- **Duan, Y.**, Wang, S. J., & Ji, Y. (2021). Hi3+3: A model-assisted dose-finding design borrowing historical data. *Contemporary Clinical Trials*, 106437.

WORKING PAPERS

- **Duan, Y.**, Guo, X., Zhong, W. Synthesis of multiple endpoints by Bayesian multi-state modeling in oncology trials. (*Manuscript in preparation*).
- **Duan, Y.**, Korpela, E., & Xu, Y. Regimen aware analysis of CNS-active medication effects on cognition in people with HIV. (*Manuscript in preparation*).

BOOKS

- Schwob, M., **Duan, Y.**, Cantoni, B., Flores-López, B., & Walker S. (2025). *Exercises in statistical reasoning*, Chapman & Hall/CRC.
- Zhuo, J., **Duan, Y.**, & Jiang X. (2022). *MATLAB operation research*, Tsinghua University Press.

AWARDS & HONORS

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|--|------------------------------|
| • University Graduate Continuing Fellowship, UT Austin | 2024-2025 |
| • Second Place Poster Award of 2025 Best of Statistical Science Workshop | Apr 2025 |
| • Winner of 2024 ASA-SBSS student paper competition | Aug 2024 |
| • Winner of the 5 th Stat4Onc Annual Symposium poster competition | May 2022 |
| • Travel Funding Award | May 2022; May 2023 |
| • Freshman scholarship and Scholarship of Fudan University | Dec 2017; Oct 2018; Oct 2019 |

TEACHING EXPERIENCE

Johns Hopkins University

Guest Lecturer

- EN.553.734 Generative AI and Agentic Systems for Clinical Data Science and Biopharmaceutical Statistics (*graduate level*), Fall 2026.

University of Texas at Austin

Teaching Assistant

- SDS 320E Elements of Statistics, Fall 2021, Spring 2024.
- SDS 322E Elements of Data Science, Spring 2022.

INVITED TALKS

- Frontiers of Bayesian Inference and Data Science, Sep 2024, Oaxaca, Mexico.
Self-supervised learning with Gaussian Processes
- Interpretable Inference via Principled BNP Approaches in Biomedical Research and Beyond, Jul 2024, Singapore.

Spatially aligned random partitions for tumor-immune interaction

- 2023 WNAR/IMS Annual Meeting, June 2023, Anchorage, AK.

Flexible evaluation of surrogate markers with Bayesian Model Averaging

CONTRIBUTED TALKS & POSTER PRESENTATIONS

- Joint Statistical Meeting, Aug 2026, Boston, MA.

Self-supervised representation learning for longitudinal polypharmacy patterns

- 2025 Best of Statistical Science Workshop (Poster Session), Apr 2025, College Station, TX.

Spatially aligned random partitions on spatial transcriptomics data

- Joint Statistical Meeting (Student Paper Competition), Aug 2024, Portland, OR.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- 2023 Leading Edge of Cancer Research Symposium (Poster Session), Nov 2023, Houston, TX.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- Women in Machine Learning (Poster Session), Nov 2022, New Orleans, LA.

Spatial clustering with random partitions on ovarian cancer data

- Conference on advances in Data Science (Poster Session), Oct 2022, College Station, TX.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- New England Statistical Society Symposium, May 2022, Storrs, CT.

A Unified Decision Framework for Phase I Dose-Finding Designs

- Stat4Onc symposium (Poster Session), May 2022, online.

A Unified Decision Framework for Phase I Dose-Finding Designs

- Society for Clinical Trials Annual Meeting, April 2022, online.

A Unified Decision Framework for Phase I Dose-Finding Designs

SOFTWARE

- **SARP**: R implementation of spatially aligned random partition model on spatial transcriptomics data.
<https://github.com/YunshanDYS/SARP>
- **SASC**: R implementation of dependent mixture models for immune profiling among colorectal cancer subtypes.
<https://github.com/YunshanDYS/SASCcode>
- **SurrogateBMA**: R package for flexible evaluation of surrogate markers with Bayesian model averaging.
<https://cran.r-project.org/web/packages/SurrogateBMA/index.html>

PROFESSIONAL SERVICE

Conference Service

- Organizer, ENAR 2027 Spring Meeting Invited Session, “*Statistical Representation Learning for Biomedical Discovery from Complex Health Data*”.
- Scientific Committee for the Bayesian Young Statisticians Meeting (BAYSM) 2025.
- Organizer, 2023 WNAR/IMS Meeting Invited Session, “*Novel methods to identify & use surrogate markers in clinical trials*”.

Journal reviewer

- Biometrics, Statistical Science, Biostatistics, The New England Journal of Statistics in Data Science, BMC Research Notes, Journal of Classification.

INDUSTRY & RESEARCH EXPERIENCE

MD Anderson Cancer Center

June 2025 – July 2025

Research Intern

Mentor: Dr. Wenyi Wang

- Developed Bayesian methods for spatial transcriptomics with applications to characterizing spatial patterns, cellular heterogeneity and interactions in the tumor microenvironment.
- Collaborated with biomedical researchers on the analysis and interpretation of spatial omics data.

Pfizer

May 2023 – Aug 2023

Biostatistics Intern

Mentor: Dr. Wei Zhong, Dr. Xiaohan Guo

- Investigated synthesis of multiple endpoints in oncology trials using Bayesian multi-state models.
- Developed a Bayesian model allowing for joint inference of overall survival, response, and progression-free survival.

Cytel

Aug 2020 – Aug 2021

Biostatistics Intern

Mentor: Dr. Yuan Ji

- Developed statistical engines using C++ and R. Implemented trial designs, debugged, and validated results.
- Developed R package/ R shiny page for novel Bayesian clinical trial designs.
- Conducted clinical trial consulting. Provided clinical trials conducting suggestions, simulation results, edited protocols.